

Hazem Alsagheer . Jarvis Consulting

I am a Computer Engineering graduate from Toronto Metropolitan University with experience building data-driven systems at the intersection of analytics and infrastructure. During my time at Hydro One, I developed Python and SQL pipelines to process and validate enterprise data, identifying critical inconsistencies that helped prevent potential regulatory losses. I have also worked in a research setting applying topic modeling techniques to large-scale datasets, where I focused on improving clustering quality through better model evaluation strategies. More recently, I gained hands-on experience with Databricks by building an end-to-end data engineering project using PySpark and the Medallion Architecture, while also earning the Databricks Data Engineer Associate certification.

Skills

Proficient: Python, RDBMS/SQL, Spark, Databricks, ETL Pipeline, Data Modeling, pandas, NumPy, FastAPI, Data Structures & Algorithms, Git

Competent: Docker, Cloud (AWS, Azure, GCP), Machine Learning, JavaScript, TypeScript, React.js, Java

Familiar: Linux/Bash, Nginx, C/C++, VBA, Networking

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_HazemAlsagheer

Linux Cluster Monitoring System [GitHub]: Built a lightweight Linux cluster monitoring system where each node collects hardware specifications and real-time resource metrics and sends them to a centralized PostgreSQL database. Designed the system to give administrators clear visibility into CPU, memory, and disk usage for better resource allocation and capacity planning. Implemented a Bash-based agent using tools like vmstat and df, with data ingestion handled through a containerized database deployed via Docker. Automated data collection with cron and validated accuracy through script testing and SQL-based analysis.

Java Grep App [GitHub]: Implemented a Java-based Grep application that recursively scans directory structures and extracts file content matching user-defined regular expression patterns, mimicking core Linux grep functionality. Developed two separate implementations using both imperative programming and Java Streams/lambda expressions to compare readability, maintainability, and performance tradeoffs under different workloads. Utilized Maven for dependency management and packaging, JUnit for automated testing, SLF4J/Logback for application logging, and Docker for containerization and deployment portability.

Python Data Analytics [GitHub]: Developed an end-to-end data analytics solution using Python, Pandas, NumPy, and Matplotlib to analyze over 1M retail transactions for a UK-based online retailer. Performed data cleaning, transformation, customer segmentation using RFM analysis, and exploratory data analysis to uncover customer retention patterns, seasonal sales trends, and geographic revenue distribution. Delivered actionable business recommendations supported by data visualizations and statistical insights, demonstrating the ability to translate raw data into strategic business decisions.

Databricks Data Engineering Project [GitHub]: Designed and implemented an end-to-end data engineering pipeline in Databricks using PySpark and the Medallion Architecture (Bronze, Silver, Gold) to process transactional and customer data from multiple sources. Built scalable ETL pipelines to ingest, cleanse, transform, and model data into Delta tables optimized for analytics. Leveraged Spark SQL and PySpark DataFrame transformations for data validation, aggregations, window functions, and business metric calculations. Delivered analytics-ready datasets and interactive dashboards that identified fraud patterns, customer behavior trends, and key business insights.

Highlighted Projects

Journaling app: Echos of Ink [GitHub]: Designed and deployed a full-stack journaling application with a focus on reliable data flow and secure system architecture. Built a FastAPI backend with JWT-based authentication and refresh token rotation, paired with a React TypeScript frontend using Axios interceptors to manage concurrent requests and session state. Modeled and managed relational data in PostgreSQL, including token lifecycle handling for security and consistency. Containerized the application with Docker and deployed it on AWS EC2 behind an Nginx reverse proxy with HTTPS, ensuring scalable and production-like behavior across environments.

Capstone Project: Biomedical Question Answering Model: Built a biomedical question-answering system by fine-tuning BioBERT on 3K+ BioASQ list-type questions, improving F1-score from 3% to 53.7% through a custom multi-span extraction strategy. Designed a post-processing pipeline to rank and filter top-N predictions, reducing redundancy

and improving answer precision. Focused on optimizing the end-to-end data and inference pipeline to ensure accurate, structured outputs for domain-specific queries.

Professional Experiences

Data Engineer, Jarvis (2026-present): Working in a fast-paced, sprint-driven engineering environment focused on building scalable software and data-driven solutions across multiple technologies and domains. Collaborate with peers through daily standups, technical discussions, and project reviews while consistently delivering projects under tight deadlines. Regularly apply problem-solving, debugging, and system design skills in an iterative development setting, adapting quickly to new tools, frameworks, and engineering concepts while communicating technical decisions clearly.

Data Analyst, Hydro One Inc. (2023-2024): Designed an SQL and Python-based ETL pipeline to spot and resolve data inconsistencies across enterprise systems, saving the company a potential regulatory fine of \$1M/day. Wrote complex SQL queries to analyze large datasets and generate actionable insights, enabling stakeholders to make faster, data-driven decisions. Resolved data quality issues at the source across cross-functional systems, improving overall data reliability by 66%. Mentored incoming co-op students on SQL development, system workflows, and best practices, strengthening team efficiency.

Research Assistant, Toronto Metropolitan University: Cybersecurity Research Lab (2024-2025): Conducted end-to-end analysis of a BERTopic pipeline on large-scale cybersecurity corpora, evaluating embedding performance and topic coherence. Identified inefficiencies in the default topic selection process, which relied on static parameter choices and led to suboptimal clustering. Designed an interval-based optimization strategy to iteratively evaluate topic counts and cluster configurations, improving coherence and interpretability. Presented findings to senior researchers, contributing to refinement of the modeling approach.

Education

Toronto Metropolitan University (2020-2025), Bachelors of Engineering, Electrical and Computer Engineering - Dean's List (2020 - 2025) - GPA: 4.06/4.33

Miscellaneous

- Databricks Certified Data Engineer Associate.
- Piano has been a passion of mine for over 10 years, including performing in concerts and largely teaching myself along the way.
- Chess enthusiast (1730 Elo) who has competed in regional and provincial tournaments.
- Train Muay Thai and calisthenics regularly, with extensive sparring experience and a focus on continuous improvement.
- Enjoy staying active through soccer and rock climbing, having climbed up to V5 bouldering routes.